

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – Summer 2022

Course: B. Tech.

Branch : Mech Engg

Semester :VII

Subject Code & Name: Industrial Engineering & Management BTMEC704B

Max Marks: 60

Date: 24/08/2022

Duration: 3.45 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Answer Any Two of the following questions

- A) Define Management. What are the basic steps in scientific management as suggested by F. W. Taylor? **CO1 6**
- B) Explain “Unity of Command” as the ‘principle of management’ suggested by Henry Fayol. **CO1 6**
- C) Explain Porter’s five competitive forces by considering an example of Toothpaste Manufacturing Company. **CO2 6**

Q.2 Attempt Any One of the following.

- A) Explain Porter’s five competitive forces by considering an example of Toothpaste Manufacturing Company. **CO1 12**
- B) Define ‘authority’ & explain the four levels of authority. **CO3 12**

Q. 3 Write short notes on Any Two of the following.

- A) Facility Layout **CO2 6**
- B) Capacity Planning **CO2 6**
- C) Product design process. **CO2 6**

Q.4 Solve Any one of the following.

- A) If surface finishing of part involves following steps, prepare flow process chart (material type) **CO4 12**
- i) Parts were brought from store 20m away from machine (1 min)
 - ii) Waiting for operator to be free from present work (20 min)
 - iii) Part loaded on the machine (3 min)
 - iv) One surface was machined (3 min)
 - v) Part was unloaded from machine (1 min)
 - vi) Activities from step ii to step v were repeated till the component was machined on all four sides.
 - vii) Do the inspection of part 25m away from machine
 - viii) Check the accuracy of surface finishing (5 min)
 - ix) Place the parts in rack for further use (1 min).
- B) Study the following case study and answer the questions following it. **CO4 12**

Jeevak Pharmaceutical Co. Ltd.

Jeevak Pharmaceutical Co. Ltd. was started as a Private Limited

Company by three technocrats having specialized training in pharmacy and chemistry at Jalgaon to manufacture and market pharmaceutical formulations – branded tablets, capsules, ointments, oral liquids some five years back. With expansion in sales, the company went public after first three years of operations and is now a company with turnover of Rs. 25 crores, a field-force of 250 representatives, some 300 workers, and a dealer network of 60 authorized stockiest all over the country. The company is poised for further growth by entering bulk drug production consisting of anti – TB and anti-diarrhea drugs. The company has adopted product management type of staff function in its marketing set up.

Questions

- i) How would you appreciate the decision of the company to introduce ‘product management type’ of structure?
- ii) Suggest the suitable type of organization chart for this company which has gone into substantial diversification.

Q. 5 Solve Any Two of the following.

- | | | |
|---|-----|---|
| A) Explain ergonomically correct “work place layout”. | CO2 | 6 |
| B) Explain Producibility & Manufacturability in concurrent engineering. | CO2 | 6 |
| C) Explain Therblings | CO2 | 6 |

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch : Mechanical

Semester :VII

Subject Code & Name: BTMEC704B Industrial Engineering and Management.

Max Marks: 60

Date:07/02/2023

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Define management. Explain in brief functions of management.	CO1	6
B) What are the aims of Scientific management? Explain any three in detail.	CO1	6
C) Explain the importance of “planning” for higher productivity.	CO4	6
Q.2 Solve Any Two of the following.		12
A) Explain in detail the objectives of Human resource management.	CO1	6
B) What is operation scheduling? Explain forward scheduling in detail.	CO3	6
C) Why preventive control is recommended over direct (Overall) control?	CO3	6
Q.3 Solve Any Two of the following.		12
A) Explain in brief Flexible manufacturing system.	CO2	6
B) Write a note on operation planning and control.	CO1	6
C) What are the Characteristics of Service System?	CO3	6
Q.4 Solve Any Two of the following.		12
A) What is capacity planning? What are the benefits of capacity planning?	CO4	6
B) What are therbligs? Give any five therbligs with symbols.	CO2	6
C) Explain in detail “method study” procedure.	CO1	6
Q.5 Solve Any Two of the following.		12
A) What is ergonomics? Write the principles of ergonomics.	CO1	6
B) Write a note on Just in Time (JIT).	CO1	6
C) List down the various steps in conducting a stop watch time study.	CO1	6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch: Mechanical Engg.& Allied Semester : VII

Subject Code & Name: BTHM702 & Industrial Engineering and Management

Max Marks: 60

Date:04-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following	12
A) What is Industrial Engineering? Discuss the role of an Industrial Engineer.	6
B) Define management. What are the various functions of management?	6
C) Differentiate between line organization and staff organization.	6
Q. 2 Solve Any Two of the following	12
A) Explain Apprentice Act 1961 along with benefit and limitations of Apprenticeship Training.	6
B) What do you understand by merit rating? Explain the objective of merit rating in detail.	6
C) What is motivation? How does it help in improving the productivity?	6
Q. 3 Solve Any Two of the following	12
A) What are the different types of manufacturing systems? Explain in detail with the help of neat sketch.	6
B) Explain forecasting? What are the different forecasting methods?	6
C) Explain Material Requirement Planning (MRP) with the help of neat sketch.	6
Q. 4 Solve Any Two of the following	12
A) What is the significance of capacity planning? Enlist the various capacity planning activities.	6
B) What are the different types of facility layout? Explain with the help of neat sketch.	6
C) What are the different factors need to be taken into account while selecting the location of plant?	6
Q. 5 Solve Any Two of the following	12
A) What is method study? What are its objectives? How does it differ from work measurement?	6
B) Explain the process of operation of the quality circle.	6
C) What is ergonomics? Enlist and explain the principles of ergonomics.	6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary End Semester Examination – Winter 2023

Course: B. Tech. Branch: Mechanical & Allied Engineering Semester: VII

Subject Code & Name: BTMEC704B-Industrial Engineering & Management

Max Marks: 60

Date: 09/01/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Answer <u>Any Two</u> of the following questions. (6x2=12)		12
A) Explain the functions of management.	CO1	
B) Shortly explain objectives and strategies of planning.	CO1	
C) Explain the following concepts of organizing, 1. Departmentation 2. Decentralization	CO1	
Q. 2 Answer <u>Any Two</u> of the following questions. (6x2=12)		12
A) Explain the importance of incentive schemes for performing staffing function of management.	CO1	
B) Explain the leadership functions of management.	CO1	
C) What is the difference between overall and preventive control of management.	CO1	
Q. 3 Answer <u>Any Two</u> of the following questions. (6x2=12)		12
A) What are the types and characteristics of services systems?	CO2	
B) Briefly explain materials requirement planning.	CO2	
C) What is assembly line balancing? Explain with suitable example.	CO2	
Q. 4 Answer <u>Any Two</u> of the following questions. (6x2=12)		12
A) Explain method study and work study in short.	CO3	
B) Explain stopwatch time study method of work measurement.	CO3	
C) Explain work sampling technique of work measurement.	CO3	
Q. 5 Answer <u>Any Two</u> of the following questions. (6x2=12)		12
A) What is ergonomics and explain its basic principles.	CO4	
B) Briefly explain concurrent engineering.	CO4	
C) Write a short note on Just in time (JIT).	CO4	

***** End *****